lumina-strobe — Design Document

PCB-0006-A

hwde v1 pipeline

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1 Board and Run Metadata

Field	Value
board	lumina-strobe
part number	PCB-0006-A
workspace	`boards/PCB-0006-A_lumina-strobe`
phase	P4
gate status	no gates recorded yet
state created	2026-07-28T14:59:51
state updated	2026-08-07T01:58:03

2 Overview

3 Requirements

Requirements: LUM-DTR-STROBE-A (LUMINA strobe daughter board)

Sources, in precedence order (later overrides earlier where they conflict):

1. `brief/00-lumina-system-context.md` - shared system context. 2. `brief/01-carrier-board-brief.md` - the mating carrier. CONTEXT ONLY, not this board. 3. `brief/02-strobe-daughter-brief.md` - this board's brief. 4. `brief/05-lumina-closed-decisions.md` - BINDING. Supersedes `00` section 6 and corrects several figures in `00` section 5 and `02` section 3. 5. `brief/06-connector-icd.md` - the frozen expansion connector ICD. A HARD input. Never redefined by this board. Per its own preamble, where a number appears in more than one place the ICD is the one that governs.

Unstated low-risk items are marked `ASSUMED:` inline. Section 9 holds every design-changing or safety-relevant unknown. D-01, D-02 and D-03 are CLOSED and are recorded here as settled requirements, not questions.

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1. Function

A fixture-specific daughter board that stacks above the universal LUMINA carrier (LUM-CAR-A) on 11.0 mm standoffs and mates through the frozen 38-position expansion connector. It contains the entire light engine for a strobe fixture: a local energy store on the 48 V raw PD rail, an inrush limiter, a mandatory bleed path, a drive stage that dumps that store into a series white LED string in short high-current bursts with hard optical edges, sense outputs (bank voltage, LED temperature) back to the carrier's ADCs, an over-temperature shutdown that acts independently of firmware, and the board-ID mechanism that lets one firmware image recognise this as a strobe.

It carries no MCU, no network interface, no power conversion from mains, and no external connector of any kind. Strobe pattern generation, the average-energy governor and colour algorithms are host/firmware, not hardware - the hardware's job is to expose what the governor needs and to fail safe without it.

Quantity: 4-6 boards, part of a first build of 8-12 fixtures.

The board exists to solve one problem: turn a ˜8.5 W sustained power budget into short, violent bursts of light with sub-millisecond edges, and degrade gracefully (dimmer flashes, never missed flashes, never a reset) when the music asks for more than the rail can deliver.

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2. Interfaces

All external electrical connection is through two board-to-board sockets on the **bottom** side of this board. **No other connector may be added** (ICD s9): no barrel jack, no DMX, no second Ethernet, no USB. The only permitted additional wiring is internal to the enclosure (the LED string - see open question 4).

2.1 J3 - POWER block

- Part: **CONNFLY DS1023-2*7SF11**, 14 positions, 2.54 mm, 600 V, 3.0 A/contact, 8.5 mm body.
- Orientation: **faces downward** - reverse-mounted THT on the bottom side, or a bottom-side SMD equivalent (ICD s7.3). See open question 6.
- Position 1 silkscreen-marked with a triangle.
- Nominal body extent **(14, 68) - (34, 78)**; position 1 at (15.3, 69.3), long axis along +x. **Provisional until the end of the carrier's P6** (ICD s7.2) - see open question 9.

Col	Row A	net	Row B	net
1	1	‘+48V_SW‘	2	‘GND‘
2	3	‘+48V_SW‘	4	‘GND‘
3	5	‘+48V_SW‘	6	‘GND‘
4	7	‘GND‘ (guard column)	8	‘GND‘
5	9	‘+12V‘	10	‘GND‘
6	11	‘+12V‘	12	‘+3V3‘
7	13	‘GND‘	14	‘+3V3‘

Design properties this map is built to have, each checkable by eye (ICD s3.1): the 48 V group is at one end and bounded by GND on every side; **column 4 is an all-GND guard column**; every supply pin has an adjacent GND; rail order along the connector is **48 -> GND -> 12 -> 3.3**, so no single-position mis-seat can put a higher rail on a lower rail's pin.

2.2 J4 - SIGNAL block

- Part: **CONNFLY DS1023-2*12SF11**, 24 positions, 2.54 mm, 600 V, 3.0 A/contact, 8.5 mm body.
- Orientation, marking and provisional-coordinate status: as J3.
- Nominal body extent (56, 68) - (88, 78); position 1 at (57.3, 69.3), long axis along +x.

Col	Row A	net	Row B	net
1	1	‘PWM0‘	2	‘PWM1‘
2	3	‘GND‘	4	‘GND‘
3	5	‘PWM2‘	6	‘PWM3‘
4	7	‘PWM4‘	8	‘PWM5‘
5	9	‘GND‘	10	‘GND‘
6	11	‘PWM6‘	12	‘PWM7‘
7	13	‘GND‘	14	‘DSPI_SCK‘
8	15	‘DSPI_MOSI‘	16	‘DSPI_MISO‘
9	17	‘DSPI_CSn‘	18	‘I2C_SCL‘
10	19	‘I2C_SDA‘	20	‘ADC0‘
11	21	‘ADC1‘	22	‘ID_ADC‘
12	23	‘ENABLE‘	24	‘FAULT‘

No 48 V exists anywhere on J4.

2.3 Signal electrical contract (ICD s3.3) - binding

Signal	Direction	Rules this board must honour
‘PWM0..7‘	carrier -> daughter	3.3 V push-pull CMOS. Default **13-bit at 9.766 kHz**. ‘PWM0-3‘ = LEDC timer 0, ‘PWM4-7‘ = LEDC timer 1. Channels on one timer share frequency AND resolution (CAR-REQ-11). Max 8 channels total (CAR-REQ-10)
‘DSPI_*‘	bidirectional	SPI mode 0, MSB first, <= 26 MHz. This board drives no line except MISO, and only while ‘DSPI_CSn‘ is low. Shared bus, one CS
‘I2C_SCL/SDA‘	bidirectional	Open drain, 400 kHz. **Pull-ups are on the carrier (4.7 k). This board must NOT fit its own**
‘ADC0‘, ‘ADC1‘	daughter -> carrier	Analogue 0-3.3 V. **Source impedance <= 10 kohm**
‘ID_ADC‘	daughter -> carrier	Carrier fits the top leg (10 k to +3V3); **this board fits the bottom leg to GND**. Code allocated by the carrier owner - see open question 8
‘ENABLE‘	carrier -> daughter	Push-pull CMOS, **active HIGH**. This board fits its own **100 kohm pull-down**, gates **every** output stage with it (including the cap-bank charge path), and **never latches it locally**
‘FAULT‘	daughter -> carrier	**Open drain, active low**, 10 k pull-up on the carrier, wire-OR’d with the carrier’s eFuse fault. **Never drive it high**

2.4 Carrier-facing functional requirements (from 02 section 6)

- **STR-REQ-17** - PWM channel count within the 8-channel ceiling. White-only needs 1-2; RGBW needs 4-5. Resolved by open question 1.
- **STR-REQ-18** - report bank voltage to a carrier ADC pin (feeds the governor, STR-REQ-06).
- **STR-REQ-19** - report LED thermistor temperature to a carrier ADC or I2C pin.

- **STR-REQ-20** - assert FAULT on over-temperature and shut down **independently of firmware**. Do not rely on the network or the MCU to prevent a thermal event.
- **STR-REQ-21** - honour the fail-safe ENABLE: outputs off with ENABLE de-asserted, including during MCU reset and firmware update.
- **STR-REQ-22** - populate the board-ID mechanism (CAR-REQ-07).

`ASSUMED:` bank voltage on `ADC0`, LED thermistor on `ADC1`, board ID on `ID_ADC`. If open question 1 closes as RGBW and per-channel sense is wanted, the sense budget re-opens and the extra channels must go on I2C.

`ASSUMED:` no SPI device on this board; `DSPI_*` left unconnected at the socket. Revisit only if open question 1 closes as RGBW and the PWM budget forces an external LED driver.

2.5 LED string interface

- Series LED string driven from the 48 V bank (STR-REQ-12): ~2.6 A at roughly 38 V string voltage, versus ~8.3 A at 12 V. This is a closed consequence of D-02.
- Whether the emitters sit on this PCB or on a remote heatsink wired to it is **not settled**
  - see open question 4. Either way the string, its wiring and its heatsink are at PoE potential (section 8).

—

3. Power

3.1 What the connector delivers - ICD s6.2, binding

```
| Rail | Sustained (af) | Sustained (at) | Peak, ms-scale, from local bulk |
Hardware fault ceiling | Connector pin capacity |
|---|---|---|---|---|---|
| `+48V_SW` | **0.25 A** | 0.50 A | 1.0 A (eFuse limit, until it latches) | 1.0 A,
**latch off** | 5.40 A |
| `+12V` | 0.75 A | 1.25 A | 2.0 A | 2.0 A converter, OCP | 3.60 A |
| `+3V3` | 0.25 A | 0.25 A | 0.50 A | 1.0 A converter | 3.60 A |
| **TOTAL, all three rails** | **8.5 W** | **18.5 W** | - | PSE overload timer
~50-75 ms | - |
```

The per-rail ceilings do not add up and are not meant to. The TOTAL is what binds. `05` states the closed-decision budget as 8.6-9.3 W (af) / 18.7-20.0 W (at) after 2.4/3.7 W carrier overhead; the ICD's 8.5 W / 18.5 W governs where they differ. Design against **af**.

Superseded, do not use: the "~10 W regulated / ~8.5 W to the light engine" chain in `00` section 5.1 and `02` section 3.1 derives from a Skyworks Si3402-B PD part this system does not use. The carrier uses a TPS2378-class PD controller plus a 100 V buck (`05`).

D-01 closed: 802.3**at** power stage, **af** classification for the first build (resistor change + a PoE+ switch upgrades it, no respin). **Daughter budgets are designed against af.** Nothing on this board may become the part that pins the fixture to Type 1.

ICD s6.3: take power on `+48V_SW`, not `+12V`, wherever possible - it skips the 48->12 conversion and is worth 0.67 W (af) / 1.30 W (at) of extra delivered power, and it removes the carrier's only real hot spot.

`ASSUMED:` this board's own housekeeping (gate drive bias, comparator, thermistor bias) is a small fraction of the budget and is taken from `+3V3` and/or `+12V`; all LED energy comes from the 48 V bank.

3.2 Energy store - D-02 closed **The energy store runs off the 48 V raw rail from the connector.** Not 33,000 uF at 12 V.

```
| Item | Value | Source |
|---|---|---|
| Bank capacitance | **~2,800 uF** | D-02 closed, ICD s6.4 |
| Capacitor voltage rating | **>= 100 V** (63 V is NOT enough at 57 V worst
case once ceramic DC-bias derating applies) | ICD s5.4 |
| Energy over the 48 -> 40 V window | **0.99 J** | ICD s6.4 |
| Energy at full 0 -> 48 V charge | **3.23 J** | ICD s6.4 |
| Max full-window flash rate the rail sustains | **8.6 Hz (af)** / 18.8 Hz (at)
| ICD s6.4 |
| Energy per flash at SYS-REQ-03's 25 Hz ceiling | **0.34 J (af)** / 0.74 J (at)
| ICD s6.4 |
| Bank droop per flash at 25 Hz | 48 -> 45.4 V (af) / 48 -> 42.1 V (at) | ICD
s6.4 |
| Cold-start charge time at the ICD sustained limit | **0.54 s (af)** / 0.27 s
(at) | ICD s6.4 |
```

SYS-REQ-03's 1-25 Hz range is reachable on af, at reduced per-flash energy above ~8.6 Hz.

- **STR-REQ-08** - select capacitors on **pulse ripple current and ESR**, not capacitance alone. A high-ESR bank sags harder than the ideal energy calculation predicts and softens the flash edge. **Ripple current rating must appear explicitly in the BOM.**
- **STR-REQ-09 / CAR-REQ-14** - inrush limiting is **this board's** responsibility (NTC or soft-start MOSFET). **Size it against the PD's 1.0 A operating current limit, not against the connector's 5.4 A rating** (ICD s8.2). Sizing against the connector is the classic way to trip the PD's 800 us foldback deglitch and brown out the entire fixture. The carrier's eFuse dV/dT is deliberately set fast and its limit sits above this board's inrush level so the two soft-starts do not fight.
- **STR-REQ-10 / CAR-REQ-17** - **bleed path across the bank is mandatory.** The carrier bleeds its own side through 100 kohm but **deliberately fits no series diode on `+48V_SW`**, so this board's bleed path is not stranded above the carrier's.

3.3 802.3 compliance clause - ICD s8.3, binding IEEE 802.3 caps PD port capacitance at roughly 180 uF; this board's ~2,800 uF is 15x that. The carrier's 48 V load switch is therefore a **compliance part**, off through detection, classification, inrush and the 80 ms window, closing only after firmware asserts ENABLE.

1. **'+48V_SW' is DEAD at power-up and stays dead for hundreds of milliseconds.** Design for it. 2. **This board must provide NO path that energises its bank from `+12V` or `+3V3` while `+48V_SW` is off.** Doing so re-creates the compliance problem behind the switch's back. 3. **No mate sequencing exists.** A dual-row header has no first-mate/last-mate control, so this board must tolerate 48 V arriving **before or after** 3.3 V, in either order.

3.4 The governor, and a derived tension worth reading

- **STR-REQ-06** - firmware implements a leaky-bucket average-energy governor; the hardware's obligation is to expose bank rail voltage to the carrier ADC.
- **STR-REQ-07** - rail sag must degrade **gracefully**: dimmer flashes, never missed flashes or a reset. A dropped beat is more visible than a slightly dimmer one.

Derived arithmetic from the closed bank size (this is the substance of open question 2):

Flash duration	Energy available (0.99 J bank, full window)	Implied drive power	Implied string current at ~38 V
10 ms	0.99 J	~99 W	~2.6 A (the closed figure)
50 ms (STR-REQ-03)	0.99 J	~20 W	~0.52 A
100 ms (STR-REQ-01)	0.99 J + rail	~11-15 W	~0.3-0.4 A
200 ms (STR-REQ-01)	0.99 J + rail	~7-11 W	~0.2-0.3 A

At the closed 2.6 A drive current the bank holds full output for **~8.6 ms** (`dt = C*dV/I = 2800 uF * 8 V / 2.6 A`), not the 100-200 ms STR-REQ-01 asks for. The "100 W strobe" in Project Plan v1.0 and the worked example in 02 section 3.2 are a **10 ms** operating point. **STR-REQ-01's 100-200 ms full-output flash and the closed bank size are not consistent at 100 W** - a 150 ms flash lands near 15 W, roughly 6x below the 2.6 A peak the drive stage is built for. This is not a blocker (the drive stage is still sized for 2.6 A peak for short accents) but it fixes what "full output" means optically, and therefore fixes the LED choice. Put to the human as open question 2.

—

4. Environment

Item	Value	Source
Enclosure	**Plastic / 3D-printed, non-conductive.** Sealed (no forced airflow)	H1-Q5, ICD s9
Internal air temperature	**56 C (af) / 69 C (at)**	ICD s7.6
Connector contact rating	-40 to +105 C	ICD s1
Room	~5 x 7 x 2.5 m basement/garage, 8-12 fixtures	‘00‘ section 1
Ingress / vibration	Not stated	-

- The 56-69 C internal ambient is a **capacitor selection constraint**: aluminium electrolytic life halves per 10 C, and the bank is the board's largest and most lifetime-sensitive part.
- **STR-REQ-15** - thermal path sized for the **sustained** average, with junction temperature during peak pulses checked against the pulsed derating curve. **Both cases must pass.**
- **STR-REQ-05** - survive 8-16 bars of maximum-rate flashing, then return to normal. `ASSUMED:` at typical tempos this is ~20-35 s. **Duration is not actually a free variable** for the thermal steady state: the rail caps sustained input at 8.5 W (af) regardless of how long the drop runs, so the steady-state thermal case is bounded by the rail, not by the bar count.
- `ASSUMED:` indoor, dry, benign. No ingress rating, no vibration or shock requirement, no condensation. Low risk - the fixture shares an enclosure with a PoE carrier already designed for the same conditions.

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5. Size and mounting

Common LUMINA footprint (ICD s7.1) - **inherited, not chosen by this board**:

Item	Value
Board outline	**100.0 x 80.0 mm**
Corner radius	**3.0 mm**, all four corners
Board thickness	**1.6 mm**
Mounting holes	**4x M3 (3.2 mm) at 5 mm inset** (a 90 x 70 mm hole rectangle) **plus a 5th M3 at (46, 74)**
Coordinate origin	board top-left corner, x right, y down
Stack arrangement	**Stacked mezzanine, daughter above the carrier**

```
| Mated board-to-board height | **11.0 mm** hard-seated, against a positive
mechanical stop |
| Standoffs | 5x M3 female-female, 11.0 mm |
| Socket orientation | **Faces downward** |
```

Per MECH-01, `board_init.py` needs `--corner-radius 3` explicitly (default is 0) and `--mounting-holes 4`; the radius is clamped to the mounting-hole inset (`margin / 2`), so 3.0 mm works at the default `--margin 6`. Read `corner_radius` and `worker_notes` in the board_init report - do not assume the requested value was honoured. H5 at (46, 74) is added at P4 as a `MountingHole_3.2mm_M3` symbol so it carries a refdes.

Per MECH-02, **ai-ee has no outline-shrink step - the P5 outline is final.** The outline, radius and hole pattern above are already closed by the ICD, so there is nothing to negotiate here; but the notch below must be in the P5 outline, not retrofitted.

5.1 The RJ45 notch - HARD requirement from P2 onward A 30 x 26 mm relief in the TOP edge, region (6, 0) - (36, 26). The outline rectangle, corner radius and 5-hole pattern are unchanged; only this local relief differs.

Two independent reasons, both load-bearing:

1. The carrier's board-edge magjack is ~15 mm tall and the stack is 11.0 mm, so the jack protrudes ~4 mm above this board's underside. Without the notch the boards **cannot be forced flat**. 2. It is the **primary reverse-insertion interlock** (CAR-REQ-16). MECH-01's 4x M3 pattern is rotationally symmetric, so a daughter can be bolted down rotated 180 degrees; rotated, the notch lands at the bottom edge and the board presents solid material over the jack. This is a mechanical stop, not a warning.

5.2 Exclusion zones - ICD s7.6, all board-relative

```
| Zone | Region | Requirement |
|---|---|---|
| **DC-DC hot zone** | **(2, 46) - (36, 68)** | **No LED drivers and no
aluminium electrolytics.** The carrier's 48->12 converter dissipates up to 1.25
W directly below in a sealed box.  This is a vertical keepout, not an in-plane
separation rule |
| **Antenna column** | **(88, 25) - (100, 55)** | **No copper on ANY layer** (no
traces, no pour, no plane) **and no metal component.** The carrier's ESP32-S3
PCB antenna is directly below; a ground plane 11 mm above detunes it.  Wi-Fi is
a supported control path per H1-Q8, so this zone is live, not void |
| **Recovery header** | **(76, 0) - (98, 20)** | Keep clear enough that a 6-way
jumper lead can be attached with this board fitted |
```

Note the interaction: the **DC-DC hot zone forbids aluminium electrolytics** exactly where a large bank would otherwise like to live, and the **notch removes 780 mm2** from the top edge, and the **antenna column removes a 12 x 30 mm strip** of all-layer copper from the right edge. The 2,800 uF / 100 V bank must be placed around all three.

Reverse-insertion proofing also relies on: different position counts at fixed asymmetric coordinates (a 2x7 socket cannot mate a 2x12 header); 180-degree rotation aligning neither connector; the 5th mounting hole breaking the hole pattern's rotational symmetry; and silkscreen - **a pin-1 triangle at position 1 of both blocks, plus a ‘^^ RJ45‘ edge arrow** matching the carrier's.

5.3 Height

- Bottom side: the 8.5 mm socket bodies inside the 11.0 mm stack. `ASSUMED:` the two sockets are the only bottom-side parts; anything else must clear the carrier's top-side components in the same footprint region.
- Top side: **unknown.** The 100 V bank is the tallest thing on this board and radial parts in this class are tens of mm tall. See open question 5.

—

6. Quantity and budget

- **4-6 strobe daughter boards**, of 8-12 total fixtures in the first build.
- Also in the program: the universal carrier (8-12), the RGBW par daughter (6-8). **The UV bar is out of scope - D-03 closed as "no board"; allocate it no connector budget, no PWM channels, no power budget, no enclosure provision.**
- System budget: **$500-1000** for fixtures + switch + enclosures (`00` section 1). A managed 8-16 port PoE switch takes a substantial share of that, so the per-fixture allowance is tight once the carrier, the daughter, the LED and its heatsink are counted.
- The ICD references a **"$30/board target"** (s1.1, in rejecting a $15.30 mated connector pair). Whether that figure is the carrier's or applies to this board is not stated. See open question 7.

—

7. Assembly

- Vendor: **JLCPCB PCBA**.
- `ASSUMED`: **single-sided (top) SMD assembly**, plus THT for the two connectors.
- The complication: the ICD requires the sockets to **face downward** - "a reverse-mounted THT part on the daughter's bottom side, or a bottom-side SMD equivalent." Reverse-mounted THT is not a standard JLC PCBA process, and a bottom-side SMD socket would make the assembly double-sided. **Open question 6.**
- Related: if the bank is built from radial aluminium electrolytics, those are THT parts too, which lands in the same process question.
- 4-6 boards is a hand-finishable quantity, so hand-soldering two connectors per board is a live option rather than a fallback.

—

8. Compliance and safety flags

Five apply. None can be guessed; the ones that need a human answer are in section 9.

8.1 48 V domain - exceeds 30 V

- Worst case **57 V DC** (IEEE 802.3 PSE maximum). Below the IEC 62368-1 **ES1** limit of 60 V DC, so **functional insulation only** - no basic/supplementary/reinforced safeguard, and no safety-mandated creepage.
- **0.60 mm minimum outer-layer copper-to-copper spacing around every 48 V net, board-wide, from the connector pads to the cap bank** (IPC-2221B Table 6-1 column B2, 51-100 V band). **This is NOT inherited automatically - this board's DRC must be set up for it.**
- Inner layers: IPC gives 0.10 mm, below JLC's 0.127 mm minimum, so the fab minimum dominates and the HV requirement is free there. The clearance applies **through the board too** - an inner-layer or opposite-face signal passing under a 48 V antipad needs the same.

- **The 0.13 mm "permanent polymer coating" column (B4) is NOT claimed.** Standard LPI soldermask is not a qualified conformal coating, and `check_creepage.py` implements only the uncoated columns - a layout designed to 0.13 mm fails P8 with no waiver mechanism.
- **Any resistor across the 48 V domain must be 0805 or larger** (0402/0603 are typically 50-75 V working) or split into two in series. **This bites the mandatory bleed resistor and the 48 V rail-sense divider** - which also has to present <= 10 kohm to the carrier ADC.
- **Capacitors on the 48 V domain must be 100 V rated.**

8.2 High pulse current - ~2.6 A LED drive Series-string drive at ~2.6 A peak from a ~48 V bank (STR-REQ-12). Below the 3 A continuous threshold, but pulsed and with deliberately fast edges, so it is flagged: trace width and copper weight, gate drive, current-limit accuracy and switching loop area all have to be sized against the pulse, not the average. The superseded 12 V topology would have been 8.3 A; avoiding that is the whole point of D-02.

STR-REQ-13 - verify the candidate LED's **pulsed** forward current derating curve, not its DC maximum. The operating point is short high-current pulses at low duty cycle.

8.3 Stored energy hazard - up to ~3.23 J

- ~2,800 uF charged to 48 V holds **3.23 J**, and it holds it **with the cable unplugged**.
- **Bleed path is mandatory** (STR-REQ-10, CAR-REQ-17), and the carrier fits no series diode on `+48V_SW`, so this board's bleed path is not stranded above the carrier's.
- The board must be **safe to handle during assembly and service**. Silkscreen warning and a defined bleed time constant are part of the deliverable.
- A short across a charged bank is a large, fast fault current independent of any rail limit.

8.4 Non-isolated topology floating at PoE potential The entire fixture is non-isolated and floats at PoE potential. Every one of these is load-bearing and lands on this board as much as on the carrier:

- **Non-conductive enclosure**, no chassis earth, no earthed mounting hardware bonded to board GND.
- **Ethernet is the only external connection. This board may not add an external connector of any kind.**
- **This board, its drive stage, its LED string and its LED wiring are all at PoE potential.** If the LED module sits on a separate heatsink, **that heatsink and its wiring are at PoE potential too.** The heatsink must not be user-accessible (H1-Q5) and must not share a mount with anything earthed - if it is touchable, metal and earthed, the non-isolated topology is non-conformant. (ICD notes this as carrier `decisions.md` OPEN-C, unresolved there; H1-Q5 closes it for this program as "plastic enclosure, heatsink enclosed".)
- **Bench hazard.** An earthed scope probe or a non-isolated USB-UART adapter ties the floating PoE return to earth. Beyond shock and damage risk, the resulting ground currents **break PD signature detection outright** (detection currents are only a few hundred microamps). **Every test point on this board carries the same silkscreen warning the carrier's recovery header does.**
- **No MOV-to-earth surge network.** An unearthed PD does not need one; do not copy one out of a reference design.

8.5 Thermal - independent of firmware **STR-REQ-20** - over-temperature must shut the output down and assert FAULT **without the MCU and without the network**. ˜8.5 W sustained inside a sealed plastic box at 56-69 C internal air, with the LED junction also seeing peak pulses, is the case that has to fail safe on its own.

8.6 Not applicable No mains voltage. No battery of any chemistry, no charging circuit. No motors. No RF transmitter on this board (the radio is on the carrier; this board's only RF obligation is the antenna column keepout in section 5.2).

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9. Open questions

Nine. Questions 1-4 change what gets designed; 5-9 are narrower but still block a decision. Defaults are offered where a sensible one exists - answering "default" to any of them is a valid answer.

—

1. White-only or RGBW? (D-04 - this board owns it)
Does the strobe fire only white, or can it also fire saturated colour?

- **White-only** covers the dominant use: rage trap and hard French rap both call for white blasts, and SYS-REQ-07 wants clean neutral white rather than RGB-mixed white. Needs 1-2 PWM channels and one drive stage.
- **RGBW** adds the coloured blasts P8 (UK bass) asks for - "all fixtures fire red or green at 80-100%". Needs 4-5 PWM channels, four drive stages, four times the LED selection and thermal work, and it splits the same 0.99 J bank four ways.

Recommendation: white-only. The one tradeoff that matters: the 6-8 RGBW pars already produce coloured light, but they cannot strobe as hard as this board can - so choosing white-only means coloured *blasts* are as fast as the pars allow, not as fast as the strobe allows. If the coloured-blast moments in P8 are meant to be percussive rather than atmospheric, that is the argument for RGBW.

Default if you have no strong feeling: white-only.

—

2. How much light per flash, and what is the fastest sustained flash rate the fixture promises?
Currently assumed 1 J per flash at an unbounded rate. **Unbounded is not achievable** - the rail delivers 8.5 W and nothing stores more than the bank holds. The arithmetic (ICD s6.4):

- Bank stores **0.99 J** per full flash (48 -> 40 V).
- **8.6 Hz** is the fastest rate at which every flash can be a full 0.99 J flash.
- Above that, energy per flash must fall. At the 25 Hz ceiling (SYS-REQ-03) it is **0.34 J**.
- A PoE+ switch later (the D-01 upgrade, resistor change only) roughly doubles both: 18.8 Hz full-energy, 0.74 J at 25 Hz.

And a second half to this question that the briefs have not reconciled: STR-REQ-01 asks for a **100-200 ms** full-output flash, but 0.99 J spread over 150 ms is only about **15 W** of drive - the "100 W strobe" figure is a **10 ms** operating point, and at the closed 2.6 A drive current the bank holds full output for only ˜8.6 ms. So:

- Short accents (10-50 ms, STR-REQ-03) can be genuinely violent: ˜20-99 W.
- Long flashes (100-200 ms, STR-REQ-01) land at ˜7-15 W.

Default: accept it. Per-flash energy capped at 0.99 J; full-energy flashes up to 8.6 Hz, tapering to 0.34 J at 25 Hz, enforced by the firmware governor; "full output" for a 150 ms flash means ˜15 W of LED drive, and the LED is chosen so that this reads as bright in a 5 x 7 m room with 4-6 strobes firing together.

If you instead want a 100-200 ms flash that is blinding at 100 W, the bank has to grow by roughly 10x, which means re-opening D-02's sizing with the carrier owner - a blocking issue against LUM-CAR-A, not something this board can decide.

—

3. LED family and beam angle

No candidate has been evaluated (STR-REQ-13 to STR-REQ-16). Two parts:

- **(a) Vendor/family preference?** The project plan names CREE and Lumileds as a direction only. *Default: no preference - the architect picks on availability at JLCPCB, pulsed derating data, and neutral/cool white with no colour cast at full current (a green or pink tint at peak is a defect, STR-REQ-14).*
- **(b) Beam angle?** The room is 5 x 7 m with a **2.5 m ceiling** - low. A narrow beam lights one square metre of floor and fails the room regardless of its lumen figure (STR-REQ-16). *Default: a wide beam, roughly 60-90 degrees, with an off-the-shelf lens or reflector rather than a custom optic.*

—

4. Is the LED on this board, or on a separate heatsink wired to it?

This changes the thermal design, the board area, and whether an internal wire-to-board connection is needed. (Internal wiring is permitted - the ICD only forbids connectors that leave the enclosure.)

- **On-board**: emitters soldered to this PCB with a heatsink bolted through it. Fewer parts, no wiring - but ˜8.5 W sustained into a 1.6 mm FR4 board sandwiched 11 mm above the carrier in a sealed box is a poor thermal path, and it competes for area with the cap bank.
- **Off-board**: an LED module on its own heatsink, connected by two internal wires.

Recommendation: off-board module on its own heatsink - it is the arrangement the ICD already assumes in section 9, and it decouples the LED's thermal problem from a crowded board. Either way the heatsink is at PoE potential and must not be user-accessible.

Default: off-board, connected by a 2-pin internal wire-to-board connector or solder pads (architect's choice).

—

5. How much room is there above this board inside the enclosure?

The 2,800 uF / 100 V bank is the tallest part on the board and parts in that class run to tens of millimetres. The enclosure is not yet designed, so this is a number that needs setting rather than discovering - and once P5 fixes the outline there is no shrink step.

*Default: state a ceiling of 30 mm** above the board's top face. If the intended enclosure is shallower, say so now - it forces the bank toward a larger number of shorter cans or a different capacitor technology, which changes the BOM cost and the board area.*

—

6. Does the downward-facing socket force double-sided assembly at JLCPCB?

The ICD requires the two sockets on the **bottom** side facing down. Three ways to build it:

- **(a)** Top-side SMD assembly at JLC, and hand-solder the two reverse-mounted THT sockets (and any THT electrolytics) locally. Cheapest, and 4-6 boards is a hand-finishable quantity.
- **(b)** Bottom-side SMD sockets, i.e. genuine double-sided assembly at JLC. Higher setup cost, but nothing to hand-solder.

- **(c)** Full JLC assembly including THT service.

Default: (a). *Answer changes: the board's assembly-side constraint from P4 onward, and the capacitor technology (radial THT vs SMD).*

—

7. What is the BOM cost target for one strobe daughter board, at quantity 6?

The system budget is $500-1000 for 8-12 fixtures plus a managed PoE switch plus enclosures. Once the carrier, the daughter, the LED and its heatsink are counted, the per-fixture allowance is tight, and the LED plus the 100 V bank are the two expensive items on this board. The ICD mentions a "$30/board target" but does not say whether that is the carrier's figure.

*Default: $25 per board** for the PCB and its BOM at qty 6, **excluding** the LED module and heatsink, which are budgeted separately with the fixture.*

—

8. What board-type ID code is allocated to LUM-STR-A, and does this board carry an I2C EEPROM?

The carrier fits the top leg of the ID divider (10 k to +3V3); this board fits the bottom leg to GND, and **the ICD says board-type codes are allocated by the carrier owner, not chosen by daughters**. This board cannot pick its own value. The ICD also notes the Q10 default keeps an I2C EEPROM alongside the divider, without saying whether the daughter or the carrier holds it.

*Default: request an allocation from the carrier owner and design to a placeholder value that must be confirmed before P8; **divider only, no EEPROM on this board** (an EEPROM buys per-unit calibration storage, which matters more for the colour-matched pars than for a white strobe).*

—

9. Proceed on the ICD's provisional connector coordinates, or wait for the carrier's P6?

ICD section 7.2's connector positions - J3 at (14, 68), J4 at (56, 68), H5 at (46, 74) - are explicitly **provisional until the end of the carrier's P6**, and the ICD says daughters are blocked on them. Everything else in the ICD is frozen.

Default: proceed on the provisional coordinates, and re-check them against the re-issued ICD before this board's own P6 completes. If they have moved, the fix is a `place_edit`, not a respin. Waiting for the carrier's P6 serialises the two runs for no design benefit.

—

Traceability

Requirement IDs referenced and where they land in this document:

ID	Section
SYS-REQ-01..08	1, 3.4 (behavioural spec, inherited via STR-REQ-01..05)
STR-REQ-01..05	1, 3.4, 4
STR-REQ-06, 07	2.4, 3.4
STR-REQ-08, 09, 10	3.2, 8.3
STR-REQ-11	1 (‘ASSUMED:‘ acceptance number is < 1 ms optical rise/fall per Project Plan v1.0; the brief leaves it to be defined during design)
STR-REQ-12	2.5, 8.2
STR-REQ-13..16	4, 8.2, open questions 3 and 4
STR-REQ-17..22	2.3, 2.4
CAR-REQ-07, 08, 10..17	2.3, 2.4, 3.2, 5
D-01, D-02, D-03	3.1, 3.2, 6 (all CLOSED)
D-04	open question 1 (OPEN, owned by this board)
MECH-01, MECH-02	5
ICD-01	2, 5 (hard input, never redefined)
DOC-01	design document is a required deliverable at the end of this run
GIT-01	scoped commits only - never ‘gate.py --commit‘, never ‘git add -A‘

—

10. H1 amendments - BINDING, supersede everything above

Authority: project owner, H1 checkpoint verdict, 2026-07-28. Recorded in `state.json` (`human --checkpoint 1 --status approved`). These entries supersede the corresponding text earlier in this document and in `brief/02-strobe-daughter-brief.md`.

10.1 D-04 CLOSED - RGBW, four colour channels

Open question 1 is closed as **RGBW**. This is **against this run's recommendation** of white-only, and was chosen knowingly with the full costed delta in front of the owner (+$4-6 board BOM, +$40-80 LED module, all 8 PWM channels consumed against a baseline of 3, and **no additional light** - RGBW divides the same ~6.6 W of sustained output into colours rather than adding to it). **Not to be re-litigated.**

Consequences this run now owns:

- Four drive stages, not one. Per-colour sense moves to I2C - the 2-ADC budget does not stretch to four colours.
- **Per-colour output is roughly a quarter of the white figure.** The 10,000 lm headline from the white-only baseline must not be carried forward; light numbers are re-derived per channel in `architecture/power_tree.md`.
- All 8 PWM channels are consumed across 4 LEDC timers (CAR-REQ-11). The flash gate is a 5-200 ms one-shot, **not** a 9.766 kHz duty setting, and the white-only baseline could only treat that as free because it owned LEDC timer 0 exclusively. **That assumption no longer holds and the timer allocation is re-checked and reported at H2.**

10.2 STR-REQ-01 AMENDED - dual-mode flash

> **Original text (superseded):** "STR-REQ-01 | Full-output flash, 100-200 ms, with > **instant** blackout either side. No visible decay tail, no fade-in ramp. | P1 rage trap: > the visual gap between maximum and zero is the entire effect."

Why it was amended, not met: 0.99 J of usable bank energy over 150 ms is ~15 W of drive, not the ~99 W the brief's "100 W strobe" implies. Reaching ~99 W for 150 ms needs 13.8 J, i.e. **32,500-65,000 uF**. Measured against this board: 12,000 uF alone is 5,104 mm2 (**64 % of a 100 x 80 mm board**) and 37 mm tall, and even that holds full output for only 36.9 ms. This is a consequence of the closed 8.5 W (af) budget and D-02's closed bank sizing - **physics, not a tooling limit or an oversight.**

STR-REQ-01 (amended): the fixture provides two flash modes.

```
| Mode | Duration | LED drive power | String current | Notes |
|---|---|---|---|---|
| **Headline / blast** | **8.68 ms** | **98.8 W** (white-only figure; ~1/4 per
colour) | 2.6 A | 0.858 J to the LED, 0.990 J from the bank.  This is "full
output" |
| **Long, mode 1** | 50 ms | 28.1 W | 0.74 A | bank + rail during the flash |
| **Long, mode 2** | 100 ms | 18.2 W | 0.48 A | |
| **Long, mode 3** | 150 ms | 14.9 W | 0.39 A | |
| **Long, mode 4** | 200 ms | 13.3 W | 0.35 A | max ~2.9 Hz repetition at 58 %
duty |
```

Unchanged and still binding: instant blackout either side, no visible decay tail, no fade-in ramp, and STR-REQ-11's <1 ms optical rise/fall. The amendment changes only the *amplitude available at long durations*, never the edge quality.

10.3 802.3at - this board is af-ONLY

BLOCKING-03 accepted. LUM-DTR-STROBE-A is designed, built and documented for **802.3af only**. **No board area and no cost is spent preserving an at path.**

An at build puts 4.67 W across two linear pass elements in sealed-box air; the best case is 1.91 W per D2PAK against a 1.40 W allowance - **it fails by 1.5x**. Capping the governor to keep it inside thermals yields ˜12.1 W of the 18.5 W available = **+45 % light, not the +120 % D-01 implies**. A real at build for this daughter needs an off-board or heatsinked pass element, i.e. a respin.

D-01's hedge ("resistor change plus a PoE+ switch, no board respin") continues to hold for the carrier and for the par. It does not hold for this daughter. This is a disclosure to the carrier owner, not an ICD change request - nothing this board needs from the carrier moves.

10.4 Light engine - SPECIFY, do not design

The RGBW LED module is **not designed by this run**. This run produces a specification complete enough that someone else can build the MCPCB: emitter selection, string topology, thermal path, and the board-to-module connector, written as explicit numbered acceptance criteria (`LE-xx`) in `architecture/light-engine-spec.md`. **That spec now covers four colour channels, not one.**

10.5 BLOCKING-01 (RJ45 notch) - owned by the coordinator, not this run

`board_init.py` is gaining a `--cutout` flag centrally, because both daughters need the 30 x 26 mm notch. **This run must NOT implement a workaround, must NOT shrink the outline, and must NOT hand-edit Edge.Cuts.** The notch region (6,0)-(36,26) stays a hard keepout in `architecture/constraints.json` regardless, so the board is electrically correct either way. The flag is assumed present by P5 and will be confirmed before this run reaches it.

10.6 CARRY - ICD s7.6 internal-air figures are PROVISIONAL

The par run raised a blocking issue against ICD s7.6: the internal-air figures are not self-consistent (69 C at cannot coexist with 56 C af), and an independent calculation gives **89-115 C**, i.e. the ICD is optimistic by **20-46 K**.

This board's entire thermal case is built on 56 C (af) internal air. Treat s7.6 as provisional until the carrier re-issues it. **How every margin moves at 85-90 C air rather than 69 C must be derived and reported at H2.**

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	Design baseline = white-only (D-04 recommendation), pending human verdict at H1	Brief leans white-only; RGBW coloured blasts are already covered by 6-8 RGBW pars. Architecture is built white-only with an explicit RGBW delta documented, so a human 'RGBW' verdict costs a P2 revision not a restart.
P0	BLOCKING-01: board_init.py cannot cut the mandatory 30x26 mm RJ45 notch (ICD s7.6)	board_swig.draw_outline emits only a rectangle (optionally corner-filleted) on Edge.Cuts; no script in the pipeline writes Edge.Cuts geometry after P5, and the playbook states the P5 outline is final. Surfaced at H1 with options.

Phase	What	Why
P1	P1 roster: 4 component-scouts (led-emitter, drive-stage, energy-store, protection-sense) + 1 reference-design (pulsed-led-driver) + power-architect. No research-interface-spec agent.	The only standards-bound interface is the expansion connector, and it is frozen and fully specified in brief/06-connector-icd.md (pin map, levels, currents, creepage). An interface-spec agent would only re-transcribe it.
P1	BLOCKING-02: STR-REQ-01 (100-200 ms flash at FULL output) is unachievable at any bank size that fits	0.99 J usable / 0.15 s = 6.6 W. Reaching ˜92 W for 150 ms needs 13.8 J = 32,500-65,000 uF. energy-store scout measured 12,000 uF as 5,104 mm2 (64 pct of a 100x80 board) and 37 mm tall, and even that holds full output only 36.9 ms. Physics plus the closed 8.5 W budget, not a tooling limit.
P2	Adopt connector ICD rev A2 (boards/lumina-carrier/architecture/connector-icd.md) as authoritative over the DRAFT-A copy in brief/06-connector-icd.md	Rev A2 adds s6.6 (binding bank-charging contract: current-limited soft-start, <=0.25 A af / <=0.5 A at, never >1.0 A for >162 ms, 3.2 J must land in the daughter limiter) and s8.4 (carrier guarantees 0 V at J3 when unprogrammed/crashed/brownout/reset), and raises board-wide 48 V outer-layer clearance from 0.60 to 0.635 mm.
P2	4-layer JLC04161H-3313; In1 AND In2 both solid GND (override planes_gen 4L default)	Thermal forces it: check_thermal 2L model floors at 55 C/W vs 4L 45 C/W, and the linear pass FET at 1.45 W in 56 C air fails P8 on 2L at any pour size. Supporting: 48 V routing needs 0.635 mm outer vs 0.127 mm inner, and the pulse loop return drops 1.53 mm -> 0.21 mm.
P2	Operating point: 38.0 V string at 2.6 A, 39.7 V window floor, bank ceiling 44.5 V normal / 48.0 V armed	38 V string reproduces the ICD s6.4 headline 0.99 J exactly while cutting board dissipation 2.33 -> 1.89 W (-19 pct) vs the 35.3 V alternative. The 44.5 V ceiling takes the pass FET from 1.03x its P8 thermal allowance to 1.77x margin at identical light output.
P2	Charge limit 0.20 A flat current limit, not the ICD s6.6 ceiling of 0.25 A	0.25 A x 48 V = 12.0 W instantaneous; plus the carrier's 2.4 W overhead that exceeds the 12.95 W 802.3af class envelope for far longer than the PSE 50-75 ms overload timer. 0.20 A satisfies s6.6's current-limited requirement literally and stays inside the class.
P2	LED-short single-fault answer: ratiometric Vds comparator (drain vs 0.45x bank) + NTC in the pass FET's own pour, latched until ENABLE cycles	The current loop otherwise regulates 2.6 A into a shorted string, burning 125-148 W in the FET with no self-termination. A max-on-time one-shot was rejected as primary: it bounds one pulse but not the repetitive 9.9 W mean.
P2	BLOCKING-03: 802.3at upgrade does not close thermally on this daughter	4.67 W across two linear FETs in 69 C air; best case 1.91 W per D2PAK against a 1.40 W allowance = fails 1.5x. The at governor cap is ˜12.1 W of 18.5 W = +45 pct light, not the +120 pct D-01 implies. A real at build needs an off-board or heatsinked pass element, i.e. a respin. D-01's 'resistor change, no respin' promise does not hold for LUM-DTR-STROBE-A.

Phase	What	Why
P2	D-04 CLOSED: RGBW, 4 colour channels	Project owner verdict at H1, knowingly against this run's white-only recommendation. Full costed delta (+$4-6 board BOM, +$40-80 LED module, all 8 PWM channels, no added light) was presented. Not to be re-litigated.
P2	STR-REQ-01 AMENDED to dual-mode: headline 0.858 J / 98.8 W / 8.68 ms, plus a separate long mode (50 ms @ 28.1 W, 100 ms @ 18.2 W, 150 ms @ 14.9 W)	H1 verdict. The original 100-200 ms at full output needs 32,500-65,000 uF (BLOCKING-02) and is unreachable at any bank size that fits. Amended by decision and written into requirements.md with the original text quoted.
P2	LUM-DTR-STROBE-A is 802.3af-ONLY. No board area or cost spent preserving an at path.	H1 verdict on BLOCKING-03. An at build fails the linear pass FETs by 1.5x in sealed-box air and buys only +45 pct light, not the +120 pct D-01 implies. The D-01 hedge still holds for the carrier and the par; this daughter is documented af-only.
P2	D-04 CLOSED: RGBW, 4 colour channels	Project owner verdict at H1, knowingly against this run's white-only recommendation. Full costed delta was presented. Not to be re-litigated.
P2	STR-REQ-01 AMENDED to dual-mode: headline 0.858 J / 98.8 W / 8.68 ms plus a long mode (50 ms @ 28.1 W, 100 ms @ 18.2 W, 150 ms @ 14.9 W)	H1 verdict. The original 100-200 ms at full output needs 32500-65000 uF (BLOCKING-02) and is unreachable at any bank size that fits. Written into requirements.md with the original text quoted.
P2	LUM-DTR-STROBE-A is 802.3af-ONLY. No board area or cost spent preserving an at path.	H1 verdict on BLOCKING-03. An at build fails the linear pass FETs by 1.5x in sealed-box air and buys only +45 pct light. The D-01 hedge still holds for the carrier and the par.
P2	CARRY: ICD s7.6 internal-air figures treated as PROVISIONAL (par run raised a blocking issue; 69 C at cannot coexist with 56 C af, independent calc gives 89-115 C)	This board's entire thermal case is built on 56 C (af) air. If real internal air is 85-90 C the pass-FET allowance collapses. Must be re-derived and reported at H2.
P2	BLOCKING-01 RESOLVED centrally: board_init.py –cutout X,Y,W,H exists, tested, committed 18613d3. P5 invocation: –outline 100x80 –corner-radius 3 –mounting-holes 4 –cutout 6,0,30,26	Coordinator added the flag; both daughters need the notch. Cutout coords are relative to the outline top-left. Notch must touch an edge (ours does, y=0) and must not overlap a corner radius (x=6 clears r=3). Interior windows are refused (exit 2) because geom._parse_outline returns on the first gr_rect and would silently treat a window as the board outline.
P2	P5 TRAP: read outline_origin from the board_init report, do not assume (0,0)	In fixed-outline mode the origin derives from the packed component bbox, so constraints.json keepout rects need translating into board space. Also verify corner_radius, cutouts and worker_notes, and confirm geom reads a non-empty outline of the expected area - a silent geometry failure reached P9 on the interior-window path.

Phase	What	Why
P2	Enclosure CLOSED: sealed non-metallic, LED heatsink bolted to/through the enclosure wall. Not vented.	Project owner; matches this run's Q6 recommendation and the par run's independent conclusion. Par measured 3.6-4.3 K/W internal-air-to-room, closing only because LED heat leaves through the wall rather than into the box. Both fixtures share one enclosure design; the wall-conduction path becomes an explicit LE acceptance criterion because this board's thermal case depends on it.
P2	RGBW emitter selection rule: published thermal data beats optical convenience; zero pulsed headroom carried through to colour	Par run swept 318 RGBW emitter parts: every Cree XLamp colour line is zero stock, and the one in-stock RGBW 4-in-1 above 50 mA publishes neither Rth nor Tj max - the owner rejected it in favour of RGB 3-in-1 plus a separate white emitter. This board's light engine is OFF-BOARD and specified, so it is NOT bound to LCSC stock; distributor sourcing must be stated explicitly.
P2	ICD s7.6 explained, not disputed: 56 C is the 'LED heat stays in the box' case	Applying the par's measured 3.6-4.3 K/W box figure to this fixture's two heat sources reproduces both disputed ICD numbers from one binary. Wall path works -> 1.894 W into the box -> 32-33 C at a 25 C room. Wall path fails -> 8.5 W -> 56-62 C, i.e. the ICD af figure to the degree. Design of record stays 56 C and the wall-conduction path is now a load-bearing dependency OF THIS BOARD, with worst-package margin 1.9-2.2x.
P3	BLOCKING-04 (P3-OPEN-1): the hot-swap charge controller's fault TIMER cannot survive the 87 pct-duty current-limit regime on EITHER candidate	Datasheet-verified at P3: TPS2490 TIMER charges 25 uA / sinks 2.5 uA (10:1); LM5069 charges 85 uA / sinks 2.5 uA (34:1). At 87 pct duty in regulation the timer integrates positive to its 4 V threshold regardless of C_TIMER -> LM5069-2 auto-restarts at 0.5 pct duty (bank stops charging), LM5069-1/TPS2490 latch off. No controller in the class omits the timer. Needs an architecture answer, not a part swap.
P3	BLOCKING-04 RESOLVED: charge path becomes a discrete timerless current limiter - ballast 2x39R 2512 + P-channel D2PAK + LM2904B half sensing across the ballast, ENABLE-gated, SOA covered by the NTC already in the pour	Break-even timer duty is 9.09 pct (TPS2490) / 2.86 pct (LM5069) against a design duty of 87 pct - no C_TIMER escapes it. Series-ballast-only (route a) rejected: the 25 Hz recharge swing (+/-0.045 A) is 1.6x the headroom (0.028 A), so the limiter re-enters regulation at any R. The ballast IS the sense element (3.9 V at the limit vs 3 mV offset). Dissipation re-splits 0.606 W into two 2512s / 0.215 W in the FET, deleting the charge FET from the 85-90 C failing set. BOM 20.53 -> 15.90 /board.
P3	BLOCKING-05: IRF640NS datasheet does NOT state gate/drain/source pin assignment or the tab's electrical node	Both package outlines number leads 1/2/3 and stop; the only G/D/S labels are on the pinless schematic symbol. Extractor correctly refused to fill it from part-family memory. Four pass-FET placements ride on it. Must be resolved from an authoritative source (IR AN-994 / JEDEC TO-263AB / the pulled EasyEDA symbol) before P4 wiring.

Phase	What	Why
P3	MCP23008 and SN74LVC1G3157 kept as DOCUMENTED DERATINGS, not re-sourced	Datasheets contradict the sourcing rationale: MCP23008's -E +125 C grade is specified ONLY at Vdd 4.5-5.5 V (on +3V3 it is specified to +85 C); the UMW SN74LVC1G3157 is -40..+125 C in the PDF, not -55, and is characterised only to +85 C. Both are within abs max (unspecified, not destructive) and design-of-record ambient is 56 C, which the wall-path analysis showed is conservative. Re-sourcing would re-open the POR fail-safe analysis for a 1.14 USD saving on 6 boards. Reversible; recorded for H2.
P3	IRF640NS SOA gap CLOSED by derivation: SOA is a pure thermal limit, no Spirito knee	All four published SOA lines satisfy P x Zth(j-c) = 150 C exactly against Fig 11, so the curve extends legitimately: ~151 W at 100 ms, ~150 W at 200 ms, landing on the 150 W Pd rating. Offsetting concern recorded: Fig 3 transfer curves cross at Vgs ~6.4 V / 24 A, so at 2.6 A the tempco is POSITIVE (current-hogging region) - single-die linear-mode thermal stability needs watching and paralleling pass FETs is unsafe.
P3	BLOCKING-06 FIXED: strobe kicad lib tables pointed at the REPO ROOT lib/, not the board's own	fp-lib-table and sym-lib-table used ${KIPRJMOD}/../../../lib/ (repo root) where the carrier correctly uses ${KIPRJMOD}/../lib/. The first lib_pull created a stray untracked repo-root lib/ that other boards could collide with, while boards/lumina-strobe/lib stayed empty. URIs corrected, assets moved into the board's lib/, stray dir deleted.
P3	ID_ADC code CLOSED by ICD s3.4 rev A3 (NORMATIVE): LUM-STR-A is code 1, R_ID = 2.7 kohm 1 pct	The P3 placeholder was 4.7 kohm - which is code 2, LUM-PAR-A. Left unfixed the strobe would have identified itself to carrier firmware as the RGBW par and loaded the wrong daughter profile. Adjacent-code separation is 0.353 V against +/-0.05 V of 1 pct divider error.
P3	PWM timer concern DISSOLVED by ICD s3.5 rev A3 (NORMATIVE): the carrier does NOT hard-assign LEDC timers	The rev-A2 statement that PWM0-3 sit on timer 0 and PWM4-7 on timer 1 is withdrawn - the ICD says it was wrong for the strobe specifically. Each daughter declares its own channel->timer->frequency map in its design document and carrier firmware applies it from the ID_ADC code. The GPIO/RMT one-shot flash gate is explicitly legal. A 14-bit / 4.883 kHz opt-in mode is also now offered (CR-3 granted).
P3	C:/Program Files/Git/OT_TRIP -> FAULT level translation VERIFIED against ICD s2 rev A6	ICD requires a daughter asserting FAULT to sink >= 5 mA (real load ~4.3 mA including the carrier's red fault LED, not the 0.33 mA the 10k pull-up alone implies). The added 2N7002 (Q404) is rated 115 mA continuous with Rds(on) ~1.2-2 ohm, so VOL at 5 mA is ~10 mV, and its 60 V Vds comfortably holds off a 3.3 V pull-up. Meets the ICD with large margin; the 12 V /OT_TRIP node never reaches the connector.

Phase	What	Why
P3	Design-of-record ambient moved to 70 C (ICD s7.7 rev A5 normative ceiling), NOT this fixture's computed 32-48 C	Designing to the published normative number rather than our own favourable computation. Worst package breaks even at 83.8 C = 13.8 K margin; even the wall-path-fails case (77 C) passes at 1.16x. Consequence: the P_rail governor caps are WITHDRAWN as operating constraints and kept only as contingency, returning 3-15 pct of the light. BANK_ARM momentary-only is now the board's sole thermal constraint. Cold start Tj 92-109 C against a 125 C design limit with the corrected 150 C Tjmax.
P3	Timer map: 1 LEDC timer used, 3 free. 14-bit/4.883 kHz mode DECLINED.	Rev B budgeted two timers and left none; the actual map needs one. 14-bit declined with numbers: comparator offset dominates resolution by 47x, and restoring the doubled ripple would push settling to 9.2 ms against a 2.67 ms shortest pulse.
P3	Floating-rail bias RESOLVED: 2 x 8.2 kohm = 16.4 kohm dropper	Gives 1.00 mA zener bias at 48 V minimum against worst-case 1.2 mA op-amp Iq, and 49 pct of the 0805 power rating at 57 V - fixes both levers of the earlier self-conflict at a cost of 0.7 pct of P_avail.
P3	H2-DECISION-1 (P3-OPEN-7): charge-FET SOA cannot be closed by derivation	The P x Zth(j-c) method that closed the IRF640NS does NOT extend to the IRF5210S: the IRF640NS had a DC line to extrapolate toward, the IRF5210S has none. Three mitigations already closed off - bigger ballast barred by the window arithmetic, lower limit barred by the 0.170 A sustained draw, soft-start ramp delays the peak without reducing it. Recommendation: accept plus 1000-cycle bench validation; fallback an off-catalogue P-channel with a published DC line.
P3	H2-DECISION-2 (P3-OPEN-12): HOLD the two Extended hysteresis values (820k bank UVLO, 620k bank ceiling)	No Basic E24 neighbour exists. Retargeting would spend 18 pct of the UVLO band - already the tightest and deliberately not widened - or take a 0.31 V bite out of the 8.3 V operating window, to save about 3 USD across the build. Sourcer correctly refused; confirming at H2 rather than silently.
P3	P4 POLARITY TRAPS recorded: electrolytic symbol has no +/- cue (pad 1 = POSITIVE); MMSZ zeners pin1=CATHODE while SMBJ58A TVS pin1=ANODE in the same BOM	Librarian verified from symbol pin names and footprint silk. A reversed 100 V electrolytic on a charged 48 V bank is an energetic failure, and the opposite pin-1 conventions on two diode families in one BOM is a live reversal trap. Both must be stated explicitly to every P4 sheet agent.
P3	fp_verify 6/6 effective: C23708's pad_count 3v4 AND pad_pitch 5.08v2.54 are ONE root cause (centre lead has no land), not two defects	D2PAK cropped centre lead means pads 1 and 3 sit two lead pitches apart, so nearest-neighbour pitch reads 5.08. Reconciled, not waived. C2622 IRF5210S reuses the same footprint file and the same 4=D/1=G/3=S map, so the trap covers 5 devices. Zero courtyard-less footprints across all 24.

Sheet plan

LUM-DTR-STROBE-A - hierarchical sheet plan

> **REV B - H1 REVISION, 2026-07-28.** RGBW. **The single `drive` sheet becomes FOUR sibling > sheets, one per colour** - see the decision in s0 below. Everything else keeps its rev A > structure; `conn`, `charge` and `protect` are edited, not rebuilt.

Seven child sheets under a thin stitching root. ~140 components across cleanly separable domains, so hierarchy buys parallel P4 sheet generation and keeps the 48 V energy store visually and electrically isolated from the analogue loops and the ICD boundary.

The sheet NAMES below are contractual. They appear verbatim in netlist net names (`/charge/CHG_GATE`, `/drive_w/LED_K`) and `architecture/constraints.json` references them; renaming a sheet renames nets and breaks `netlist_audit --constraints`.

0. Four sibling sheets, not one sheet instantiated four times [REV B]

Decision: four distinct sheet files - `drive_w`, `drive_r`, `drive_g`, `drive_b` - each with its own refdes band and its own `pwr_base`. Rejected: one `drive.kicad_sch` instantiated four times.

Three reasons, in the order that decides it:

1. **Refdes uniqueness is trivially guaranteed** by disjoint bands, with no dependence on KiCad's per-instance annotation data surviving a pipeline round-trip. Every P4/P8 check in this pipeline keys on refdes. 2. **Net names come out readable and distinct**: `/drive_w/LED_K` vs `/drive_r/LED_K`. A repeated instance produces path-qualified duplicates of the same local name, and `constraints.json` declares four *different* 2.6 A / 57 V nets that P8 must resolve individually. 3. **P6 wants four independent placement groups anyway** - each stage clusters tightly around its own pass FET and drain pour, and those pours must be >= 16 mm apart (`power_tree.md` s11).

Cost of the choice: four near-identical sheet files instead of one. **They must be generated from one parameterised routine at P4 so they cannot drift** - a hand-edited copy of a drive stage is exactly the kind of divergence that shows up as an intermittent single-colour fault on the bench.

```
| Sheet | File | Blocks | Interface nets (sheet pins) | refdes range | ‘pwr_base‘ |
|---|---|---|---|---|---|
| root ‘lumina-strobe‘ | ‘lumina-strobe.kicad_sch‘ | none (stitching only) | - | -
| 1 (unused) |
| **‘conn‘** | ‘conn.kicad_sch‘ | J3 POWER, J4 SIGNAL, ENABLE pull-down, ID
divider bottom leg, ADC RC filters, rail decoupling, mounting holes, test
points | ‘ENABLE‘, ‘FAULT‘, ‘PWM0‘-‘PWM7‘, ‘ADC0‘, ‘ADC1‘, ‘ID_ADC‘, ‘I2C_SCL‘,
‘I2C_SDA‘, ‘VBANK_SENSE‘ | **1-99** | **100** |
| **‘charge‘** | ‘charge.kicad_sch‘ | input TVS, hot-swap controller, charge FET,
sense shunt, 2720 uF bank, passive + active bleed, bank divider | ‘ENABLE‘,
‘VBANK‘, ‘VBANK_SENSE‘, ‘CHG_EN_n‘ | **100-199** | **200** |
| **‘drive_w‘** | ‘drive_w.kicad_sch‘ | white stage:  error amp, setpoint RC +
SPDT, pass FET, current shunt, gate clamp, drain TVS, Vds divider | ‘VBANK‘,
‘PWM0‘, ‘PWM4‘, ‘ENABLE‘, ‘OT_TRIP‘, ‘UVLO_n‘, ‘VDS_SENSE_W‘, ‘LED_K_W‘ |
**200-249** | **300** |
| **‘drive_r‘** | ‘drive_r.kicad_sch‘ | red stage, identical | ‘VBANK‘, ‘PWM1‘,
‘PWM5‘, ‘ENABLE‘, ‘OT_TRIP‘, ‘UVLO_n‘, ‘VDS_SENSE_R‘, ‘LED_K_R‘ | **250-299** |
**400** |
| **‘drive_g‘** | ‘drive_g.kicad_sch‘ | green stage, identical | ‘VBANK‘, ‘PWM2‘,
‘PWM6‘, ‘ENABLE‘, ‘OT_TRIP‘, ‘UVLO_n‘, ‘VDS_SENSE_G‘, ‘LED_K_G‘ | **300-349** |
**500** |
| **‘drive_b‘** | ‘drive_b.kicad_sch‘ | blue stage, identical | ‘VBANK‘, ‘PWM3‘,
‘PWM7‘, ‘ENABLE‘, ‘OT_TRIP‘, ‘UVLO_n‘, ‘VDS_SENSE_B‘, ‘LED_K_B‘ | **350-399** |
**600** |
| **‘protect‘** | ‘protect.kicad_sch‘ | 2x **quad** comparator on ‘+12V‘ (board
OT, LED OT, bank UVLO, bank ceiling, **4x Vds fault latch**), NTC dividers,
TMP112, **I2C I/O expander**, ‘BANK_ARM_n‘ fail-safe stage, harness connector
J200, off-board NTC landing J300 | ‘VBANK‘, ‘VBANK_SENSE‘, ‘VDS_SENSE_*‘ x4,
```

‘LED_K_*‘ x4, ‘ENABLE‘, ‘OT_TRIP‘, ‘UVLO_n‘, ‘CHG_EN_n‘, ‘FAULT‘, ‘ADC1‘, ‘I2C_SCL‘, ‘I2C_SDA‘ | **400-499** | **700** |

[REV B] ‘J200‘ moves from ‘drive‘ to ‘protect‘. With four colours it is a single 6-way connector fed by four different sheets, so it cannot live inside any one of them. `protect` is the sheet that already sees all four `LED_K_*` nets (the Vds comparators) and already owns `J300`, so it owns the harness boundary too.

Exception to the refdes ranges: `J3`, `J4` and `H5` keep the designators the ICD assigns them (ICD s7.2). Everything else follows the band scheme, which is what keeps refdes unique across sheets without coordination.

‘pwr_base‘ values are 100 apart and disjoint across all seven sheets (100 / 200 / 300 / 400 / 500 / 600 / 700), so `#PWR` designators never collide even where a component band and another sheet's power band share a decade - `R400` and `#PWR0400` are different namespaces, but two sheets both emitting `#PWR0400` would not be.

—

1. Net naming - the CANONICAL final netlist names

Mechanism (verified in this repo at `tests/s7_regen/hierdemo/kicad/hierdemo.net`):

1. **Power SYMBOLS make a net global across the whole hierarchy** with a **bare** name and need no sheet pin. 2. A child net merged with the root through a sheet pin takes the **root-side label**: `Project.add_sheet(child, ..., nets=["X"])` wires each sheet pin outward to a root local label `X`, so the net becomes **‘/X‘**. 3. A child-internal label becomes **‘/<sheet>/NAME‘**.

These are the names ‘constraints.json‘ uses and they are contractual. Getting one wrong makes `check_creepage` hard-error (it raises on a declared voltage net that is not on the board) and makes `check_current` / `check_thermal` silently no-op.

1.1 Global rails - power symbols, bare names

Net	Present on	Driven by	PWR_FLAG?
‘GND‘	all four	J3 pins 2/4/6/7/8/10/13 (passive)	**yes**, on ‘conn‘
‘+48V_SW‘	conn, charge	J3 pins 1/3/5 (passive)	**yes**, on ‘conn‘
‘+12V‘	conn, **all four drive sheets**, protect	J3 pins 9/11 (passive)	**yes**, on ‘conn‘
‘+3V3‘	conn, protect	J3 pins 12/14 (passive)	**yes**, on ‘conn‘

All four PWR_FLAGs live on `conn`, the sheet that owns the connector, at `#FLG0100+`. Power symbols being global, one flag each drives the net hierarchy-wide.

‘+48V_SW‘ deliberately does not appear on any ‘drive_ sheet or on protect‘. The only 48 V-domain net that crosses into a drive sheet is `/VBANK`, and it does so through the harness connector and the pass FET. If `+48V_SW` ever appears on a drive sheet or on `protect`, that is a design error, not a routing convenience - it would be a path that could energise something ahead of the charge FET.

1.2 Root-crossed signals - `/NAME` [REV B]

Net	Direction	Notes
‘/ENABLE‘	conn -> charge, all four drive, protect	Active HIGH. **100 k pull-down on ‘conn‘** (ICD s8.2, mandatory). Gates the charge path’s EN, **all four** pass-FET gate clamps, and the active bleed’s disarm. **Never latched locally**
‘/FAULT‘	protect -> conn	**Open drain, active low.** No pull-up on this board - the carrier’s 10 k owns it. **Never driven high.** With four stages it is a single wire-OR: **which** colour faulted is read over I2C, not from this pin

‘/PWM0‘	conn -> drive_w	**FLASH_GATE_W.** 5-200 ms one-shot at 1-25 Hz. LEDC timer 0 or GPIO/RMT - ‘blocks.md‘ s4.3
‘/PWM1‘	conn -> drive_r	**FLASH_GATE_R.** As above. **[REV B] This pin carried ‘BANK_ARM‘ in rev A; ‘BANK_ARM‘ has moved to I2C** (‘blocks.md‘ s4.4)
‘/PWM2‘	conn -> drive_g	**FLASH_GATE_G**
‘/PWM3‘	conn -> drive_b	**FLASH_GATE_B**
‘/PWM4‘	conn -> drive_w	**AMP_SET_W.** 13-bit at 9.766 kHz, LEDC timer 1, RC-filtered into the regulator reference
‘/PWM5‘	conn -> drive_r	**AMP_SET_R**
‘/PWM6‘	conn -> drive_g	**AMP_SET_G**
‘/PWM7‘	conn -> drive_b	**AMP_SET_B**
‘/ADC0‘	conn (local)	Bank voltage telemetry, from ‘/VBANK_SENSE‘ through an RC at the pin. Rth 9.43 k, inside the ICD’s 10 kohm. **Also the open-string self-test channel** (‘blocks.md‘ s4.5)
‘/ADC1‘	protect -> conn	LED-module thermistor telemetry, independent ‘+3V3‘-referenced leg
‘/ID_ADC‘	conn (local)	Board-type ID, **bottom leg only** - the carrier fits the 10 k top leg. **[REV E] ‘R_ID‘ = 2.7 kohm 1 % = ICD rev A6 s3.4 code 1 (LUM-STR-A), ‘V_ID‘ 0.702 V. ALLOCATED, not a placeholder**
‘/I2C_SCL‘, ‘/I2C_SDA‘	protect <-> conn	**Two devices: TMP112 and the I/O expander.** 400 kHz. **No pull-ups on this board** - the carrier’s 4.7 k own the bus. Addresses must not collide; both selectable at P3
‘/VBANK‘	charge -> all four drive, protect	Bank positive. **57 V, up to 10.4 A** with four colours firing. **Must be poured, not routed**
‘/VBANK_SENSE‘	charge -> conn, protect	Divided bank voltage (2 x 82 k + 10 k). Three consumers: ‘ADC0‘, the UVLO and ceiling comparators, and **all four** Vds comparators’ reference leg
‘/VDS_SENSE_W‘, ‘_R‘, ‘_G‘, ‘_B‘	drive_* -> protect	Divided pass-FET drain voltage, one per colour (2 x 82 k + 10 k each, identical ratio to the bank divider so each trip is ratiometric)
‘/LED_K_W‘, ‘_R‘, ‘_G‘, ‘_B‘	drive_* -> protect	Pass-FET drain = string cathode, one per colour. **57 V, 2.6 A each.** They cross to ‘protect‘ only because ‘J200‘ lives there (s0); the thermal pour for each belongs to its own drive sheet’s FET
‘/OT_TRIP‘	protect -> all four drive, conn	Wire-OR of **seven** open-collector outputs (board OT, LED OT, 4x Vds fault, spare). Pulls **all four** gate clamps **and** ‘FAULT‘
‘/UVLO_n‘	protect -> all four drive	Bank undervoltage lockout. Inhibits the drive stages only - it must **not** assert ‘FAULT‘ (an empty bank at power-up is not a fault)
‘/CHG_EN_n‘	protect -> charge	Bank-ceiling comparator output, pulls the hot-swap controller’s ‘EN‘ low above the ceiling

Deleted from rev A: `/VDS_SENSE` (unqualified) and `/drive/LED_K` - both are replaced by the four colour-qualified nets above. **‘BANK_ARM‘ no longer exists as a root-crossed net at all**: it is `/protect/BANK_ARM_n`, an expander output that never leaves the `protect` sheet.

1.3 Sheet-internal nets that `constraints.json` names

Net	Sheet	Why it is declared
‘/charge/CHG_GATE‘	charge	**64 V - the highest-voltage net on the board** (V_GATE-OUT is 12-16 V above VCC). Declared in ‘voltages‘ so the clearance rule covers it
‘/drive_w/LED_K‘, ‘/drive_r/LED_K‘, ‘/drive_g/LED_K‘, ‘/drive_b/LED_K‘	drive_*	**[REV B]** Pass-FET drain = string cathode. **57 V, 2.6 A each**, and the thermal-pour net for that colour’s FET. **Four independent 48 V-domain pours that also need 0.635 mm from each other**
‘/drive_w/ISNS‘, ‘/drive_r/ISNS‘, ‘/drive_g/ISNS‘, ‘/drive_b/ISNS‘	drive_*	Pass-FET source to shunt high side. **2.6 A each**, but low-voltage nodes -

deliberately **not** in ‘voltages‘, so ‘check_creepage‘ treats them as 0 V and demands the full clearance from the matching ‘LED_K‘ across the FET |
| ‘/protect/BANK_ARM_n‘ | protect | **[REV B]** I2C expander output -> bank-ceiling comparator. Active LOW to arm; 100 k to ‘+12V‘ through a 2N7002 holds it disarmed with ‘+3V3‘ dead. Never leaves the sheet, so it is not in ‘constraints.json‘ |

1.4 A merge decision worth recording **research/power.json declares /VBANK and /LED_A as separate power constraint entries. They are one net here, /VBANK.** Nothing sits between the bank's positive terminal and the harness connector's pin 1 - no fuse, no series element - so /LED_A would be an alias, and a declared net that does not exist in the netlist makes check_creepage hard-error at P8. **/LED_A is deleted, not renamed. **‘/LED_K‘ survives -** [REV B] as four nets, /drive_w/LED_K .. /drive_b/LED_K** - because there genuinely is a device (a string) between each of them and /VBANK.

—

2. Sheet contents and refdes allocation

2.1 conn - refdes 1-99, pwr_base 100

Refdes	Part	Note
‘J3‘	POWER socket, 2x7, 2.54 mm, CONNFLY DS1023-2*7SF11 class	**bottom side, faces DOWN.** Pin map = ICD s3.1 verbatim
‘J4‘	SIGNAL socket, 2x12, CONNFLY DS1023-2*12SF11 class	**bottom side, faces DOWN.** Pin map = ICD s3.2 verbatim
‘H1‘-‘H4‘	M3 3.2 mm mounting holes	generated by ‘board_init --mounting-holes 4‘, board-only
‘H5‘	‘MountingHole_3.2mm_M3‘ symbol at (46, 74)	**added at P4 as a symbol so it carries a refdes** - ‘--mounting-holes‘ makes corner holes only
‘R1‘	100 k 0603	**ENABLE pull-down, ICD-mandated**
‘R2‘	**[REV E] ‘R_ID‘ = 2.7 kohm 0603 1 %**	**ICD rev A6 s3.4 code 1 = LUM-STR-A**, ‘V_ID‘ 0.702 V. **NO LONGER A PLACEHOLDER.** The 4.7 kohm carried since rev A was **code 2 = LUM-PAR-A** - left alone this board would have announced itself as the par and been handed the wrong daughter profile
‘R3‘, ‘C3‘	ADC0 RC (1 k + 10 nF)	9.43 k source impedance is at the ICD ceiling; SAR sampling charge wants a local reservoir
‘R4‘, ‘C4‘	ADC1 RC	same
‘C1‘	‘+12V‘ local bulk, **<= 4.7 uF**	deliberately small so the error amp dies promptly on unplug
‘C2‘, ‘C5‘	‘+3V3‘ and ‘+12V‘ 100 nF	
‘TP1‘-‘TP6‘	test points: ‘+48V_SW‘, ‘/VBANK‘, ‘GND‘, ‘/OT_TRIP‘, ‘/UVLO_n‘, ‘/ENABLE‘	**every one carries the floating-PoE silkscreen warning.** **[REV B]** the per-colour ‘ISNS‘ probe points are now ‘TP<N+0>‘ on each drive sheet, so the shared ‘/drive/ISNS‘ test point is gone

2.2 charge - refdes 100-199, pwr_base 200

Refdes	Part	Note
‘D100‘	TVS, **SMBJ58A**, SMB	58 V standoff / 93.6 V clamp, under the 100 V cap rating. One part number shared with the four drain clamps
‘R100‘, ‘R101‘	**[REV C] Ballast, 2 x 39 ohm 2512 2 W in parallel = 19.5 ohm**	**BLOCKING-04.** Series ballast **and** the loop's current-sense element - 3.9 V at the 0.20 A limit. 0.303 W each. Split in two for the same reason the bleed is
‘Q100‘	**[REV C] Charge FET, D2PAK/DPAK P-channel 100-150 V**	Tab = drain = **‘/VBANK‘** (it was ‘+48V_SW‘ with the N-channel). **P-channel: no

charge pump, passively OFF.** `>= 350 mm2 F.Cu + 350 mm2 B.Cu`, sized by the **cold-start transient**, not by the 0.215 W steady row. **NOT part of the four-pass-FET consolidation** |
`U100`	**[REV C] LM2904BIDR**, SOIC-8, on a **floating 12 V rail** referenced to `+48V_SW`	A = charge-loop error amp; B = spare. **Replaces the deleted LM5069MM-2.** Same part number as the four drive-stage op-amps: a placement, not a feeder
`R102`, `R109`, `D101`, `C100`	**[REV E] Floating rail: 2 x 8.2 k 0805 IN SERIES (= 16.4 k) + 12 V zener + 1 uF**	**The rev C single 24 k was self-conflicting** - it left as little as 0.30 mA to bias the zener against the LM2904B's 1.2 mA max Iq, *and* dissipated 84 mW at 57 V = 67 % of an 0805. 16.4 k split in two gives **1.00 mA of zener bias at the 48 V minimum** and **61.7 mW per part = 49 %**. Full arithmetic in `power_tree.md` s3.3.1. Dies with `+48V_SW`, which is correct - there is nothing to charge without it
`R103`, `R104`, `D102`	**[REV C]** Source-to-gate 100 k (**default OFF - the charge path's interlock of record**), gate series, 12 V gate-source zener clamp	No POR window: the OFF state is the unpowered state
`R105`, `R106`, `C101`	**[REV C]** Reference divider off the floating rail + loop compensation	Zener-referenced, so the limit is temperature-stable to a few %
`Q101`, `R107`, `R108`	**[REV C]** 2N7002 ENABLE level shift + pull-ups	Gates the loop's reference to zero. Same pattern as the four drive-stage gate clamps
`C110`-`C113`	Bank HF: 4 x 10 uF / 100 V X7S 1210	**2.7 uF each effective at 48 V**
`C120`-`C125`	Bank bulk: **six D18 / 7.5 mm radial footprints, four populated** at 680 uF / 100 V	Second-sources the bank at 470 uF across four vendors; 2720 -> 4080 uF knob with no respin
`R110`	Passive bleed, 100 k 0805, 150 V working	23 mW, 272 s. **Un-defeatable**
`R111`, `R112`	Active bleed, 2 x 470 ohm 2512 2 W	1.23 W peak each, inside the continuous rating - **no joule rating needed, and none is published**
`Q110`	Bleed switch, 150 V N-ch SOT-223, CJT04N15 class	
`Q111`, `R113`, `D110`	Disarm 2N7002, 1 M bias from `/VBANK`, 10 V zener clamp	**Bias UP from the bank, pull DOWN to disarm** - so "every rail dead" is the ON state
`R120`, `R121`, `R122`	Bank divider, 2 x 82 k + 10 k, 0805	Rth 9.43 k. Satisfies the 0805 rule *and* the series-split rule at once

2.3 drive_w / drive_r / drive_g / drive_b - ONE TEMPLATE, FOUR INSTANCES [REV B] All four sheets are generated from a single parameterised routine at P4.

The only inputs are the colour suffix, the refdes base and the two PWM pins. Nothing else differs - not a value, not a part, not a footprint.

sheet	suffix	refdes base `N`	`pwr_base`	gate pin	amplitude pin
`drive_w`	`W`	**200**	300	`/PWM0`	`/PWM4`
`drive_r`	`R`	**250**	400	`/PWM1`	`/PWM5`
`drive_g`	`G`	**300**	500	`/PWM2`	`/PWM6`
`drive_b`	`B`	**350**	600	`/PWM3`	`/PWM7`

Refdes	Part	Note
`Q<N+0>`	**[REV C] `IRF640NSTRLPBF`**, D2PAK planar HEXFET-5, 200 V / 18 A, -55..+175 C	Tab = drain = `/drive_x/LED_K`. **>= 350 mm2 F.Cu pour + >= 350 mm2 B.Cu mirror + >= 12 thermal vias** - `power_tree.md` s5. **One part number across all four pass FETs**; `Q100` is no longer in this consolidation (it is a P-channel, s2.2)
`R<N+0>`	Shunt, 200 mohm 3 W 2512 1 %	520 mV FS. **Kelvin layout** - sense traces to the pad ends

‘U<N+0>‘	**[REV C] ‘LM2904BIDR‘**, dual op-amp SOIC-8, **on ‘+12V‘**	A = error amp / gate driver; B = setpoint buffer, free. -40..+125 C. **Same part as the charge loop’s ‘U100‘**
‘U<N+1>‘	**[REV C] ‘SN74LVC1G3157DCKR‘**, SPDT analogue switch, SC-70-6	Steers the **reference**, not the gate. **-55..+125 C in the same land - this closes rev B’s explicit P3 action item on the SGM3157** (-40..+85 C)
‘R<N+1>‘, ‘R<N+2>‘, ‘C<N+0>‘	Setpoint RC + divider: 10 k + 100 nF, 5k36/1k00 (6.35:1)	tau 1 ms, 4.6 ms to 1 %, 2.6 % ripple at 9.766 kHz
‘R<N+3>‘	Gate pull-down 4k7	**The interlock of record** - off with *everything* dead
‘R<N+4>‘	**[REV D] Gate series ‘R_g‘ = 100 ohm**	**Isolates the op-amp output from the FET’s ~2000 pF ‘Ciss‘, which is 20x the LM2904B’s rated ‘CLOAD‘.** Puts the gate pole at 796 kHz, 79x above the loop crossover. ‘blocks.md‘ s2.3.1
‘R<N+8>‘, ‘C<N+2>‘	**[REV D] Loop compensation: ‘R_in‘ = 10 k (shunt into the inverting input), ‘C_f‘ = 1 nF (output to inverting input)**	**Dominant-pole compensation - the capacitive load is INSIDE this loop by construction and cannot be isolated out of it.** Crossover 10.1 kHz, 1 % settling 73 us, total optical edge 85 us = **12x margin on STR-REQ-11** and 3.2 % of the shortest 2.67 ms pulse
‘Q<N+1>‘, ‘Q<N+2>‘	2 x 2N7002 (JLC **Basic**)	One inverts ENABLE, one clamps gate-to-source. Also the landing point for ‘/OT_TRIP‘ and ‘/UVLO_n‘
‘D<N+0>‘	Drain-source TVS, SMBJ58A class	Harness ‘L di/dt‘ clamp. **NOT a freewheel diode across the string**
‘R<N+5>‘, ‘R<N+6>‘, ‘R<N+7>‘	Vds divider, 2 x 82 k + 10 k, 0805	Identical ratio to the bank divider, so the fault trip is ratiometric
‘C<N+1>‘	‘+12V‘ decoupling, 100 nF	
‘TP<N+0>‘, ‘TP<N+1>‘	shunt Kelvin probe point, gate	short-return probe point at the shunt - a standard ground lead cannot measure this edge

Instantiated: **‘Q200‘/‘Q250‘/‘Q300‘/‘Q350‘** are the four pass FETs; R200/R250/R300/R350 the four shunts; and so on. ~18 components per sheet, 72 across the four.

2.4 protect - refdes 400-499, pwr_base 700 [REV B]

Refdes	Part	Note
‘U400‘	**Quad** comparator, LM2901 class, SOIC-14, **on ‘+12V‘**	A = board OT, B = LED-module OT, C = bank UVLO, D = bank ceiling. **LM2901 (-40..+125 C), NOT LM339/LM393 (0..+70 C)**
‘U401‘	**Quad** comparator, LM2901 class, SOIC-14, **on ‘+12V‘**	A-D = **the four per-colour Vds fault detectors**, each latched
‘RT400‘-‘RT404‘	5 x NTC 10 k 0603	**One in each of the five power-FET drain pours** (four pass + charge). **All five in parallel as one divider bottom leg** - the hottest device dominates, so a single comparator section ORs all of them. Sheet-owned here because they form one divider; **placement puts each inside a different pour** (‘constraints.json‘ groups)
‘R400‘-‘R409‘	OT dividers and ratiometric references from ‘+12V‘	supply-independent trip
‘R442‘-‘R449‘, ‘R450‘-‘R453‘	**[REV D] HYSTERESIS NETWORKS - the LM2901 has none and its datasheet offers no guidance.** Per section: ‘R_a‘/‘R_b‘ reference divider + ‘R_hyst‘ from the open-collector output back to the reference node + ‘R_pull‘ **10 k to ‘+12V‘**	**U400A board OT** 56 k / 10 k / **560 k** = 175 mV = **5.0 C** of tab; **U400B LED OT** 10 k / 10 k / **150 k** = 379 mV = **5.2 C** of solder point; **U400C UVLO** 22 k / 5.1 k / **820 k** = 59 mV = **1.02 V** of bank; **U400D ceiling** 20 k / 5.6 k / **620 k** = 82 mV = **1.42 V** of bank. Full arithmetic and the ‘Vio‘ judgement call in ‘blocks.md‘ s2.4.2. **The pull-up participates in the divider - it is not a free choice**

‘R454‘–‘R457‘	**[REV D] Vds hysteresis, ~1 M per section**	~5 % of the trip threshold. **Non-critical because these four are latched** (‘Q400‘–‘Q403‘); the band exists only to hand each latch one clean edge
‘Q404‘, ‘R458‘, ‘R459‘	**[REV D] ‘/OT_TRIP‘ -> ‘FAULT‘ open-drain translator, 2N7002**	**New, and forced by the hysteresis:** the pull-ups put ‘/OT_TRIP‘ on a **‘+12V‘** node, which the four 2N7002 gate clamps want anyway, but a 12 V node cannot land on the carrier’s 3.3 V ‘FAULT‘ pin. ‘Q404‘ translates it down as **open drain with no pull-up**, so the carrier’s 10 k still owns the net and **‘FAULT‘ is never driven high**. ‘blocks.md‘ s2.4.4
‘D400‘, ‘D401‘, ‘R460‘, ‘R461‘, ‘C414‘, ‘C415‘	**[REV D] ‘J300‘ ESD protection: bidirectional TVS (SMAJ12CA class) + 10 nF to GND per thermistor line, plus a 1 k series resistor into each comparator pin**	**The LM2901 is only 500 V HBM and ‘U400B‘’s input reaches an off-board connector.** **The 1 k must sit AFTER the divider tap, not in the loom** - in the loom it joins the NTC leg and shifts the trip point by ~1 k against a ~1 k bottom leg. After the tap it carries 25 nA of bias = 25 uV
‘R410‘–‘R425‘, ‘C400‘–‘C403‘	**4 x** Vds threshold (~0.45) + arming RC (10 k + 10 nF = 100 us blank)	one set per colour
‘Q400‘–‘Q403‘	**4 x** 2N7002 fault latch	**Cleared only by ‘ENABLE‘ going low** - fault latches, not ENABLE latches. Each latch state also goes to ‘U420‘ bits 0-3
‘R430‘–‘R433‘	Bank UVLO and ceiling thresholds, hysteresis	UVLO trip = ‘max(V_string) + 1.7 V‘ over all four colours, **set by one 1 % resistor at P3 from the measured strings**
‘U410‘	TMP112 class, SOT-563, **on ‘+3V3‘**	I2C telemetry. **Fit no pull-ups**
‘U420‘	**[REV C] ‘MCP23008T-E/SS‘**, I2C 8-bit I/O expander, SSOP-20, **on ‘+3V3‘**	bits 0-3 = four fault latch states, 4 = board OT, 5 = LED OT, **6 = ‘ARM‘ (output, active HIGH)**, 7 = spare. **Fit no pull-ups.** Address must not collide with ‘U410‘. **‘RESET‘ MUST be wired to ‘ENABLE‘, not tied to ‘+3V3‘** - see ‘blocks.md‘ s2.4.1
‘Q410‘, ‘R434‘, ‘R435‘, **‘R436‘**	‘BANK_ARM_n‘ fail-safe stage: 2N7002 + 100 k to ‘+12V‘ + gate series + **[REV C] 100 k gate-to-GND pull-down**	**[REV C] The gate pull-down is new and load-bearing**: the MCP23008 powers up with every pin as an **input (high-Z)**, not outputs-high like the PCF8574 rev B assumed, and a floating MOSFET gate is neither on nor off. With the pull-down, POR / ‘+3V3‘ dead / expander DNP all resolve to **disarmed**
‘J200‘	**[REV C] CJT ‘A3963WV-6P‘**, VH-compatible 6-pin THT, 10 A, **-40..+105 C**	2 x ‘/VBANK‘ anode (10.4 A total) + 4 x ‘/drive_*/LED_K‘ cathode (2.6 A each). **Top edge.** Pinout fixed by ‘light-engine-spec.md‘ LE-12. **The +105 C rating CLOSES ‘power_tree.md‘ s10.5’s housing question - no derating exception is needed**
‘J300‘	**[REV C] CJT ‘A2541WV-4P‘**, XH-compatible 4-pin, **-40..+105 C**	LED-module NTC (trip, ‘+12V‘) and LED-module NTC (telemetry, ‘+3V3‘). **JLC stocks no leaded/probe NTC at all** - the thermistors themselves are hand-terminated, not a PCBA line
‘R440‘, ‘R441‘	LED-module NTC divider, **NTC as the TOP leg**	an open harness wire pulls the node low and **trips** - fail-safe on a broken wire
‘C410‘–‘C413‘, **‘C416‘**	Decoupling: **100 nF X7R 0603 at each of ‘U400‘/‘U401‘ ‘V+‘, ‘U410‘, ‘U420‘, plus one shared 1 uF**	**[REV D] The LM2901 publishes no decoupling guidance and no layout section at all**, so this is a design decision ‘blocks.md‘ s2.4.3 owns explicitly - "not grounded", not "not needed". Do not leave it to a schematic agent
‘TP400‘, ‘TP401‘	‘/OT_TRIP‘, ‘/UVLO_n‘	

—

3. What P4 must review before generating [REV B]

Six items that are invisible on the bench and expensive at P8:

1. **The 48 V domain boundary.** No component may bridge `+12V` or `+3V3` to `+48V_SW`, `/VBANK` or **any** `/drive_*/LED_K` (ICD s8.3 point 2). The only nets crossing are `ENABLE/PWM` into GND-referenced inputs and the **five** divider taps, which run **out** of the 48 V domain into high-impedance inputs. Violating this re-creates the 802.3 port-capacitance problem behind the compliance switch's back and shows up only as a PD compliance failure. 2. **'FAULT' has no pull-up on this board and is never driven by a push-pull output.** Every device touching it is open collector or open drain. 3. **No I2C pull-ups**, on either device. The carrier's 4.7 k own the bus. 4. **[REV B] The four drive sheets must be byte-identical apart from suffix, refdes base and the two PWM pins.** Generate them from one routine and diff them. A hand-edited divergence in one colour is the classic source of an intermittent single-channel fault that only appears on the bench. 5. **[REV B] 'U420' bit 6 is an OUTPUT and everything else on the expander is an INPUT.** A PCF8574's quasi-bidirectional I/O powers up **high** on every pin; the design depends on that being the *disarmed* state, which is why the signal is `BANK_ARM_n` and not `BANK_ARM`. Getting the polarity backwards means the bank charges to 48 V whenever firmware is not looking. 6. **[REV B] '/VBANK' must be a pour, not a net the router is free to trace.** 10.4 A. Flag it to P6/P7 explicitly - `check_current` will fail a routed trace and the rework loop is expensive.

Full architecture narrative (not inlined; see the artifact index): `architecture/blocks.md`, `architecture/power_tree.md`, `architecture/stackup.md`.

5 Schematic

Gate `erc`: *not recorded. reports/schematic.pdf not available.*

6 Layout

Pending — produced at P6.

7 Verification

Pending — produced at P8.

8 DFM and Fabrication

Pending — produced at P9.

9 Run Record (Appendix)

Phase digests

P0

```
# P0 Intake - digest

- 'requirements-analyst' read all five brief files and wrote 'requirements.md'
(627 lines).
Both connector pin maps transcribed verbatim from the ICD; nothing invented.
- Closed decisions recorded as settled, not re-raised:  D-01 (at power stage /
af
classification, budgets against af), D-02 (~2800 uF bank on the 48 V raw rail,
100 V
parts, ~2.6 A series string, mandatory bleed), D-03 (UV bar out of scope), the
ICD rail
contract, the 802.3 port-capacitance clause, the full mechanical footprint
including the
```

```
30 x 26 mm RJ45 notch and three keepouts, and the corrected 8.5 W (af) / 18.5 W
(at)
budget.
- Safety flags raised:  48 V domain (57 V worst case), 2.6 A pulsed drive, 3.23
J stored
energy with the cable unplugged, non-isolated topology floating at PoE
potential
(LED wiring and heatsink included), sealed-box ambient 56-69 C.
- Substantive finding:  STR-REQ-01 (100-200 ms flash at FULL output) and the
closed bank
sizing are not consistent.  0.99 J over 150 ms is ~15 W of drive, not the
briefs' 100 W.
Folded into open question 2 rather than silently resolved.
- Source conflict recorded, not averaged:  `05` gives 8.6-9.3 W (af) to the
daughter, the
ICD s6.2 gives 8.5 W total.  Per the ICD's own precedence clause the ICD
governs.
- 9 open questions produced.  D-04 (white-only vs RGBW) is this board's to
close.
- Tooling gap found while checking mechanics:  `board_init.py` cannot cut the
mandatory
notch (BLOCKING-01, logged as a P0 decision).
```

P1

```
# P1 Research - digest

Roster:  4 `research-component-scout` (led-emitter, drive-stage, energy-store,
protection-sense) + 1 `research-reference-design` (pulsed-led-driver) + 1
`research-power-architect`.  No `research-interface-spec`:  the only
standards-bound
interface is the expansion connector and it is frozen and fully specified.

- **led-emitter**:  no JLC-stocked white LED is DC-rated for 2.6 A, and every
published
pulse rating in this market is at ~100 us / 10 % duty - 50x to 2000x shorter
than the
5-200 ms flashes here.  Cree's XP-G2 datasheet contains the word "pulse" zero
times.
Design rule adopted:  budget ZERO pulsed headroom, keep peak inside each die's
DC max.
Verdict:  array goes OFF-BOARD on an aluminium MCPCB + heatsink (on-board FR4
would be
120-200 C of rise).  The right optic is no optic - bare 120 deg at 2.3 m
already covers
the 5x7 m room; any TIR narrows it to the STR-REQ-16 failure mode.
- **drive-stage**:  JLC has no HV linear CC driver near 2.6 A and no wide-SOA
linear-mode
MOSFET at all, so the discrete op-amp + shunt + FET loop is forced, not
preferred.
200 mohm 2512 shunt beats 50 mohm (13 % vs 54 % offset error at STR-REQ-04's 10
% dim
point).  Found the LED-short fault:  the loop keeps regulating 2.6 A into a
shorted
string, 125-148 W in the FET, no self-termination.
- **energy-store**:  4 x 680 uF/100 V = 2720 uF, 27 mm tall, $6.18/board.
Ripple and ESR
are NOT binding (8.7x ripple margin, 1.43 % IR sag); height, endurance and
Extended
```

```
fees are.  Growing the bank 4x does not fit (64 % of the board, 37 mm tall) and
even
then holds full output only 36.9 ms.
- **refdesign-pulsed-led-driver**:  TI TIDA-01081 is direct prior art.  The
decay tail is
the driver's inductor, output capacitance and soft-start - never the LED
(phosphor
persistence ~60 ns, four orders below the 1 ms spec).  Shunt-FET dimming
REJECTED: it
keeps burning current while dark.  Linear-mode SOA rules:  pick older
generations and
higher voltage classes, never select on Rds(on).
- **power**:  zero local regulators, zero magnetics, one current limiter.
Board
dissipation is an invariant, `P_rail x (48 - V_string)/48 + housekeeping` -
string
voltage is the only first-order lever.  Exactly ONE full-energy flash from a
full bank.
And the finding that matters beyond this board:  **the 802.3at upgrade does not
close
thermally.**
```

P2

```
# P2 Architecture - digest

Adopted connector ICD **rev A2**
(`boards/lumina-carrier/architecture/connector-icd.md`)
over the DRAFT-A copy in `brief/`:  it adds s6.6 (binding current-limited
bank-charging
contract) and s8.4 (carrier guarantees 0 V at J3 when
unprogrammed/crashed/reset), and
raises board-wide 48 V outer clearance from 0.60 to 0.635 mm.

- **4 layers, `JLC04161H-3313`.** Forced by thermals, not routing:
`check_thermal`'s 2L
model floors at 55 C/W vs 4L's 45 C/W, and the linear pass FET fails P8 on 2
layers at
ANY pour size.  In1 AND In2 both solid GND - a deliberate override of
`planes_gen`'s 4L
default.  Antenna column is voided on all four layers.
- **Four sheets**:  conn (ICD boundary), charge (0.20 A hard-limited hot-swap +
2720 uF /
100 V bank + self-powered bleed), drive (op-amp + 200 mohm shunt + D2PAK planar
HEXFET,
no inductor and no output cap), protect (2x dual comparator on +12V: board OT,
LED OT,
Vds fault latch, bank UVLO + ceiling).
- **Operating point:  38.0 V string at 2.6 A, 39.7 V window floor, bank ceiling
44.5 V
normal / 48.0 V armed.** 38 V reproduces the ICD's own headline 0.99 J exactly
while
cutting board dissipation 2.33 -> 1.89 W (-19 %).  The 44.5 V ceiling is what
takes the
pass FET from 1.03x its P8 allowance to 1.77x margin, at identical light
output.
- **Charge limit 0.20 A, not the ICD's 0.25 A ceiling**:  0.25 A is 12.0 W
instantaneous
which, plus the carrier's 2.4 W, exceeds the 12.95 W af class envelope for
longer than
```

```
the PSE's 50-75 ms overload timer.  Also found:  the charge path is a
permanently-cycling
limiter in regulation ~88 % of the time, not a start-up element.
- **LED-short fault decided** (not deferred):  ratiometric Vds comparator,
drain vs
0.45 x bank off identical dividers so no voltage reference is needed, ~20 us,
arming-blanked, latched until ENABLE cycles - plus an NTC in the pass FET's own
pour.
A max-on-time one-shot was rejected as primary:  it bounds one pulse, not the
repetitive
9.9 W mean.
- **Emitter re-decided** off the false JLC constraint (the array is off-board,
so it is
not a JLC PCBA line item):  single 3S string of >=2.6 A-DC-rated multi-die
emitters
(XHP70.3 12 V class).  No parallel paths means no ballast and no Vf binning.
P3's
part-sourcer will not find it on JLC; that is correct and expected.
- **Cost ~$24.50/board at qty 6** (~$14.00 BOM + ~$3 PCB + ~$7.50 Extended
setup),
against the $25 default target.  Light engine separate at ~$46-65/fixture.
- Two things carried to H1:  **802.3at does not close thermally on this
daughter**, and
**the RJ45 notch cannot be cut by this pipeline**.
```

P3 *(no digest recorded)*

P4 *(no digest recorded)*

Gate attempt history

(no entries)

Issues

(no entries)

Human checkpoints

Id	Phase	Status	When	Note
1	P2	approved	2026-07-28T19:53:15	D-04 = RGBW (against the run recommendation, chosen knowingly). STR-REQ-01 amended to dual-mode. Strobe accepted as 802.3af-only. Light engine to be SPECIFIED not designed, now 4 channels. BLOCKING-01 owned by the coordinator: board_init –cutout flag being added centrally.
2	P4	not reached		
3	P6	not reached		
4	P8	not reached		
5	P10	not reached		

Generation cost

Not recorded (no reports/cost.json).

10 Artifact Index

File	Size	Note
state.json	33,360 B	
requirements.md	39,569 B	
architecture/blocks.md	91,265 B	
architecture/constraints.json	13,842 B	
architecture/light-engine-spec.md	27,915 B	
architecture/power_tree.md	84,383 B	
architecture/sheets.md	26,832 B	
architecture/stackup.md	24,138 B	
reports/fp_verify_C142961.json	1,774 B	
reports/fp_verify_C1849422.json	2,474 B	
reports/fp_verify_C23708.json	3,421 B	
reports/fp_verify_C2681224.json	2,072 B	
reports/fp_verify_C28927.json	3,367 B	
reports/fp_verify_C629435.json	2,233 B	